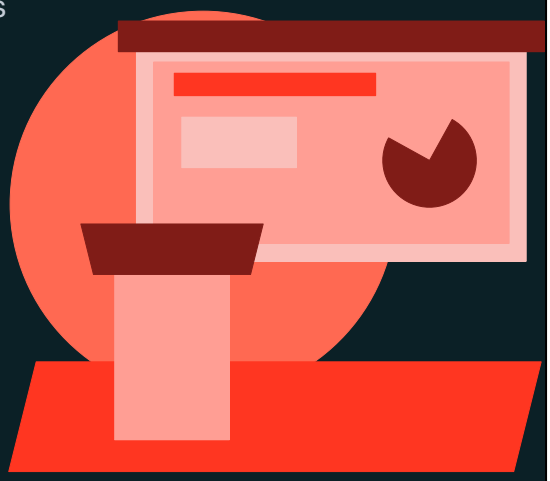
databricks

Building Lakeflow Spark Declarative Pipelines

LECTURE

Additional Features Overview



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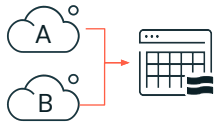
In this lecture, you will gain an understanding of advanced capabilities available in Lakeflow Declarative Pipelines, including how to use features like multi-flow pipelines, external sinks, advanced Delta Lake integrations, Unity Catalog support, and automated deployment for more flexible, scalable, and enterprise-ready data workflows.

Additional Features Overview

What's Next: Features Outside This Course



FLOWS

Add **flows** for the **same target**, creating a union of the two data sources (or tables)

SINKS

Sink API **enables writing to destinations** outside of the pipeline:

- Kafka
- Delta
- Event hubs

Full DELTA Support

Improved Delta Support for MVs/STs

- Liquid clustering
- Row-level security and column masking
- Change data feed (CDF) from STs

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While this course covers the core capabilities of Lakeflow Spark Declarative Pipelines, there are several advanced features available outside of what we've demonstrated here. These features extend your pipeline's flexibility, scalability, and integration across systems.

Flows

- You can add multiple flows that write to the same target.
- This creates a union of different data sources, allowing you to blend multiple pipelines into a single output table.
- For example, ingesting from two different customer systems into one unified customer table.

Refer to: [Load and process data incrementally with Lakeflow Declarative Pipelines flows | Databricks on Google Cloud]

Additional Features Overview

What's Next: Features Outside This Course

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Sinks

- The Sink API allows you to write pipeline outputs to external systems, not just Delta tables.
- Supported destinations include:
  - Kafka
  - Delta (within or across workspaces)
  - Azure Event Hubs
- This is great for real-time downstream consumption or system integration.
-

Refer to: [Use sinks to stream records to external services with Lakeflow Declarative Pipelines | Databricks on Google Cloud]

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Full Delta Support

Lakeflow pipelines benefit from full native Delta Lake support, which includes:

- **Liquid Clustering** - Automatically optimizes files for performance during pipeline runs.
- **Row-Level Security & Column Masking** - Enforce fine-grained access controls directly within your streaming and materialized tables.
- **Change Data Feed (CDF)** – Capture row-level changes from streaming tables and use them in downstream applications, such as audit logs or CDC propagation.

Additional Features Overview

What's Next: Features Outside This Course

Unity Catalog

Full UC Support

- Publish to multiple catalogs & schemas
- Read **STs** and **MVs** in Dedicated Access Mode
- Create a Unity Catalog pipeline by cloning a Hive metastore pipeline

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Full UC Support

- Publish to multiple catalogs & schemas (as seen in this course)
- Read STs and MVs in Dedicated Access Mode (as seen in this course)

Refer to:

- Dedicated compute group access | Databricks on AWS
- Create a Unity Catalog pipeline by cloning a Hive metastore pipeline | Databricks on AWS

Additional Features Overview

What's Next: Features Outside This Course

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Better Performance

Improved performance throughout Lakeflow Spark Declarative Pipelines!

- Serverless
- Incremental refresh for materialized views
- Photon

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Better Performance

- Improved performance throughout Lakeflow Spark Declarative Pipelines.
- Serverless
- Incremental refresh for materialized views
- Photon

Refer to:

- Configure a serverless pipeline | Databricks on AWS
- Incremental refresh for materialized views | Databricks on AWS
- What is Photon? | Databricks on AWS

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Databricks Asset Bundles (DABs)

DABs enable you to **programmatically** validate, deploy, and run Databricks resources such as pipelines for **CI/CD production workloads**

- Automated Deployment with Databricks Asset Bundles course

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Deployment with Databricks Asset Bundles (DABs)

DABs enable you to **programmatically** validate, deploy, and run Databricks resources such as pipelines for **CI/CD production workloads**
Refer to:

- What are Databricks Asset Bundles? | Databricks on AWS
- Automated Deployment with Databricks Asset Bundles | Databricks

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Thank you for completing this lesson and continuing your journey to develop your skills with us.